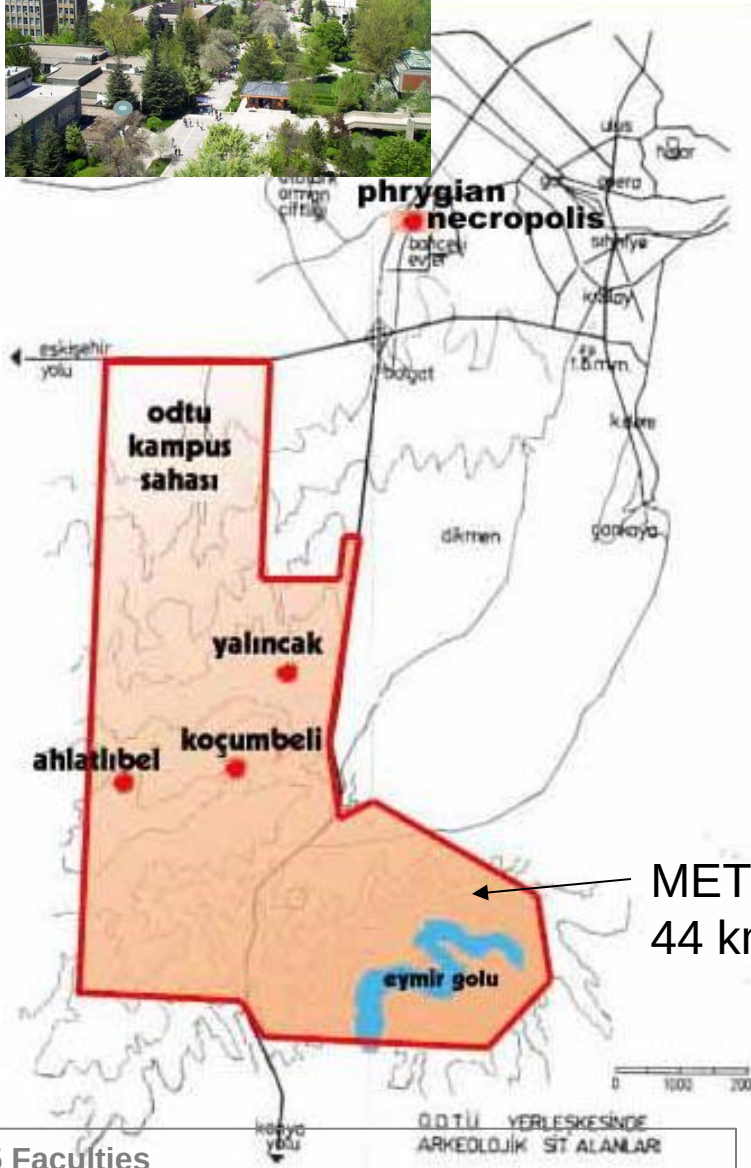


Cost-effective Processing of Hydrogen Storage Alloys from their Oxides

Serdar Tan and **Tayfur Öztürk**

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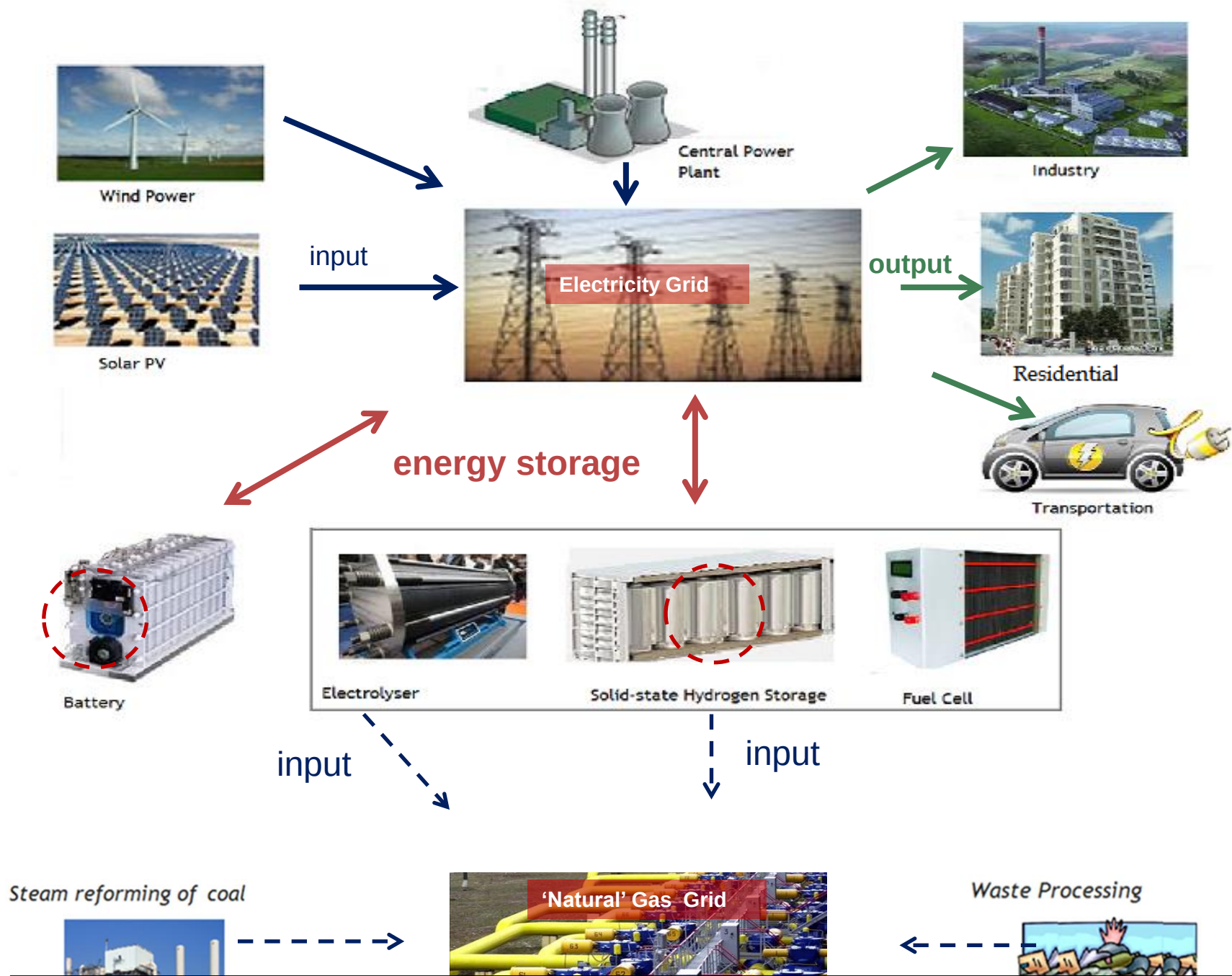
MP1103 Nanostructured materials for solid-state hydrogen storage



5 Faculties
14 Departments in Engineering
10 Departments in Art & Sciences
85 master programs ; 65 doctorate programs



15, 000 Undergraduates
5,000 MS
3000 PhD
1,500 Foreign Students



Theoretical storage capacity of some hydrogen storage alloys

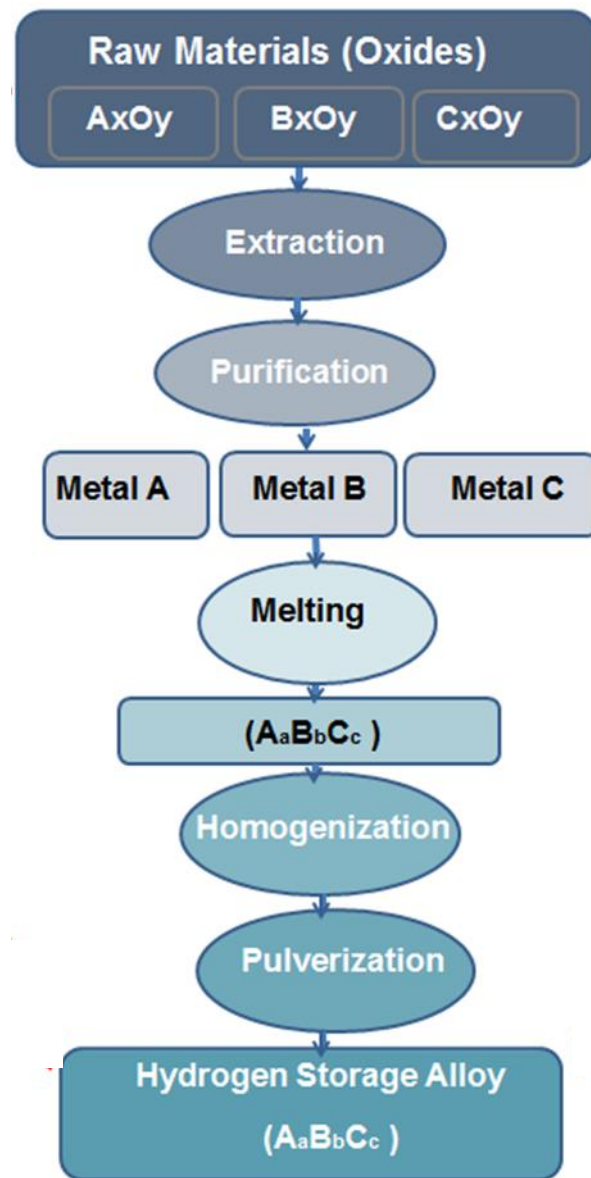
Commercial
Alloys

Current Work

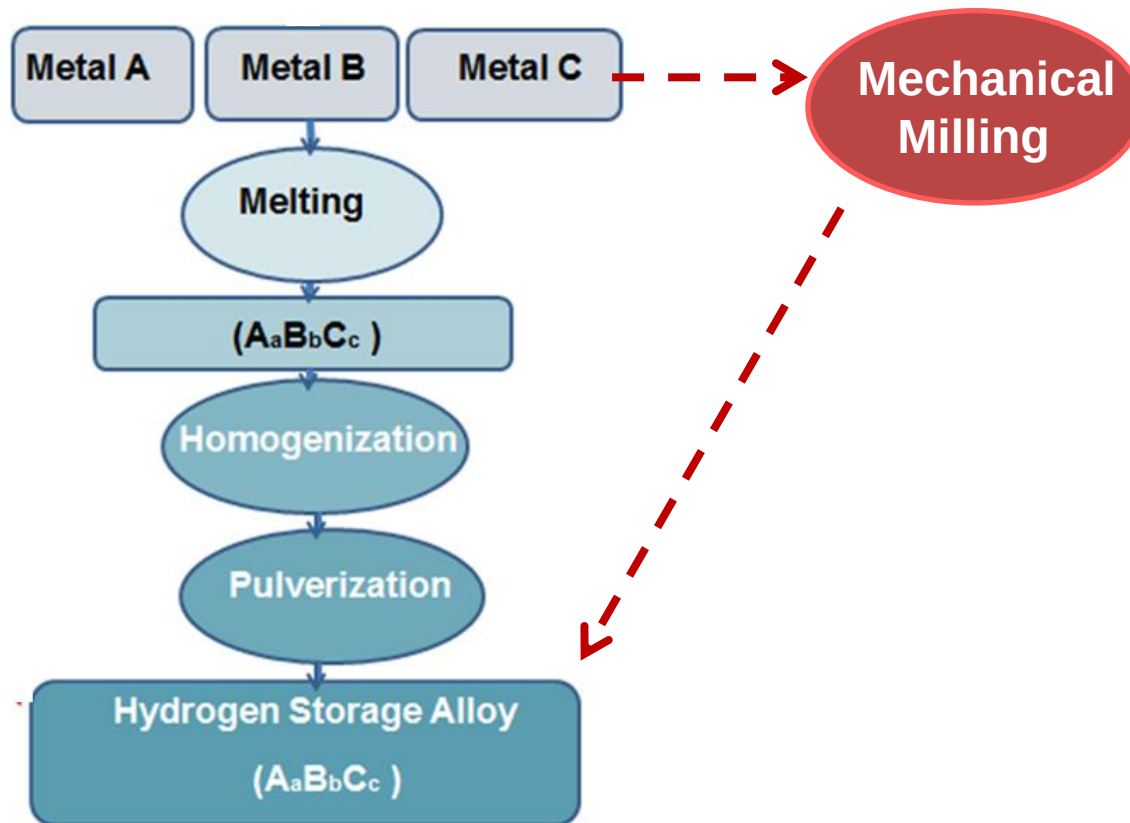
Alloy	Capacity (wt %Hydrogen)	Capacity (mAh/g)
TiNi	0.9	250
RE based AB ₅ Alloy*	1.4	372
Ti based AB ₂ Alloy*	1.5	400
Ti ₂ Ni	1.6	432
TiFe	1.9	515
ZrV ₂	2.8	750
Mg ₂ Ni	3.7	1002
Mg ₉₅ Cu ₅ **	5.5	1490

* Liu et al. J. Mater. Chem., 2011, 21, 4743

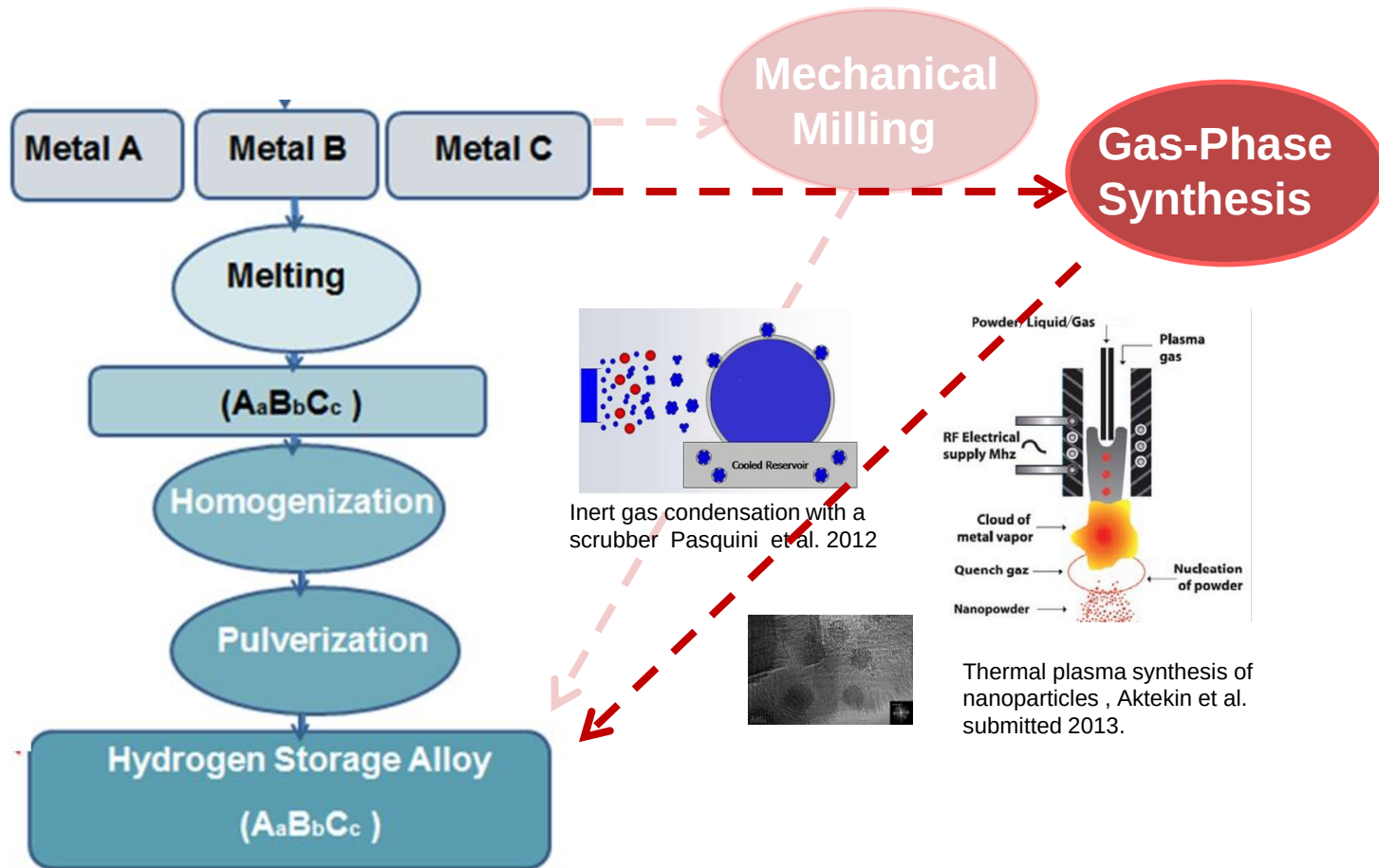
** Akyildiz & Ozturk Journal of Alloys and Compounds 492 (2010) 745–750



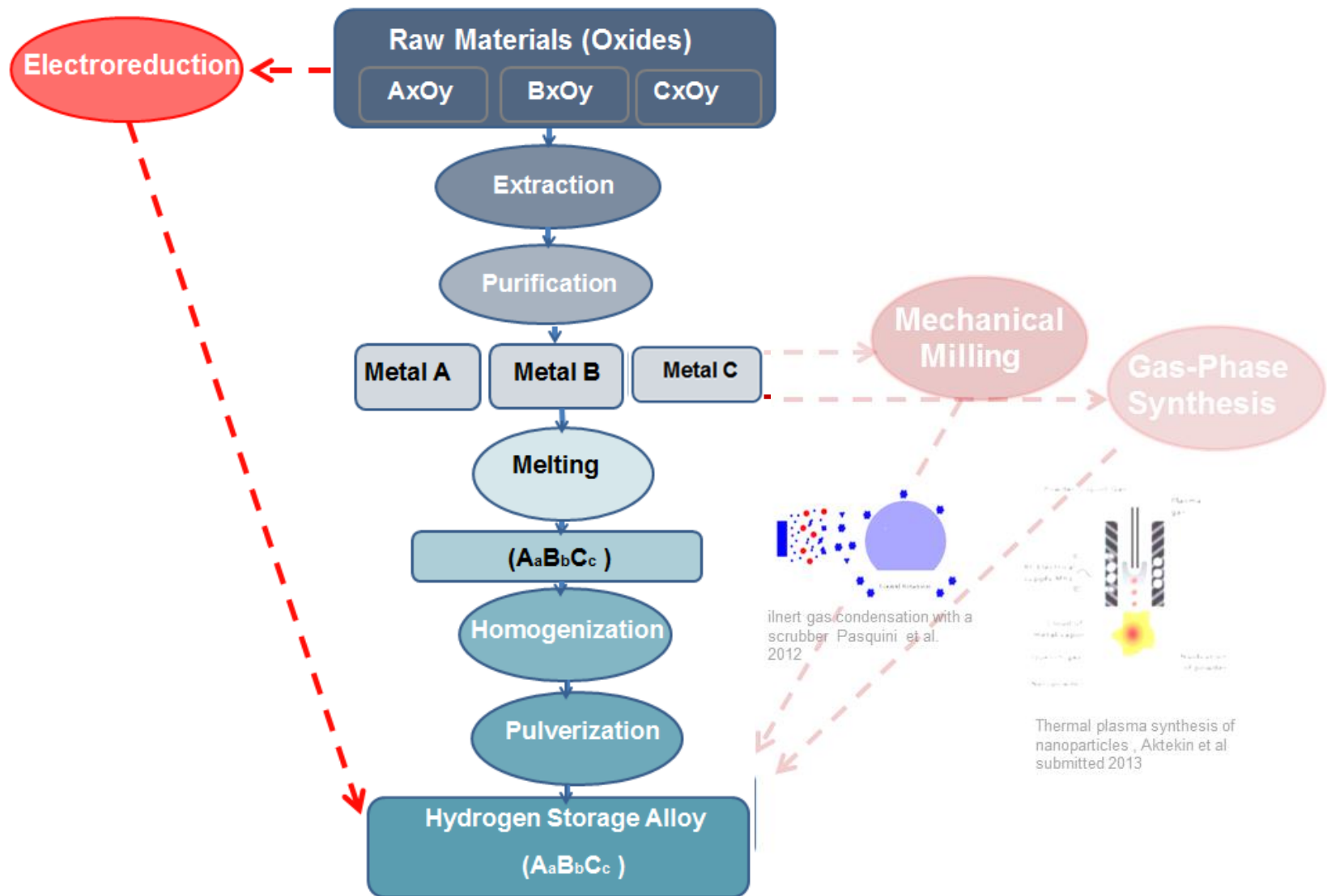
Flowchart for synthesis of hydrogen storage compounds
(Conventional Route)



Flowchart for synthesis of hydrogen storage compounds
(Nanostructured HSA via Mechanical Milling)

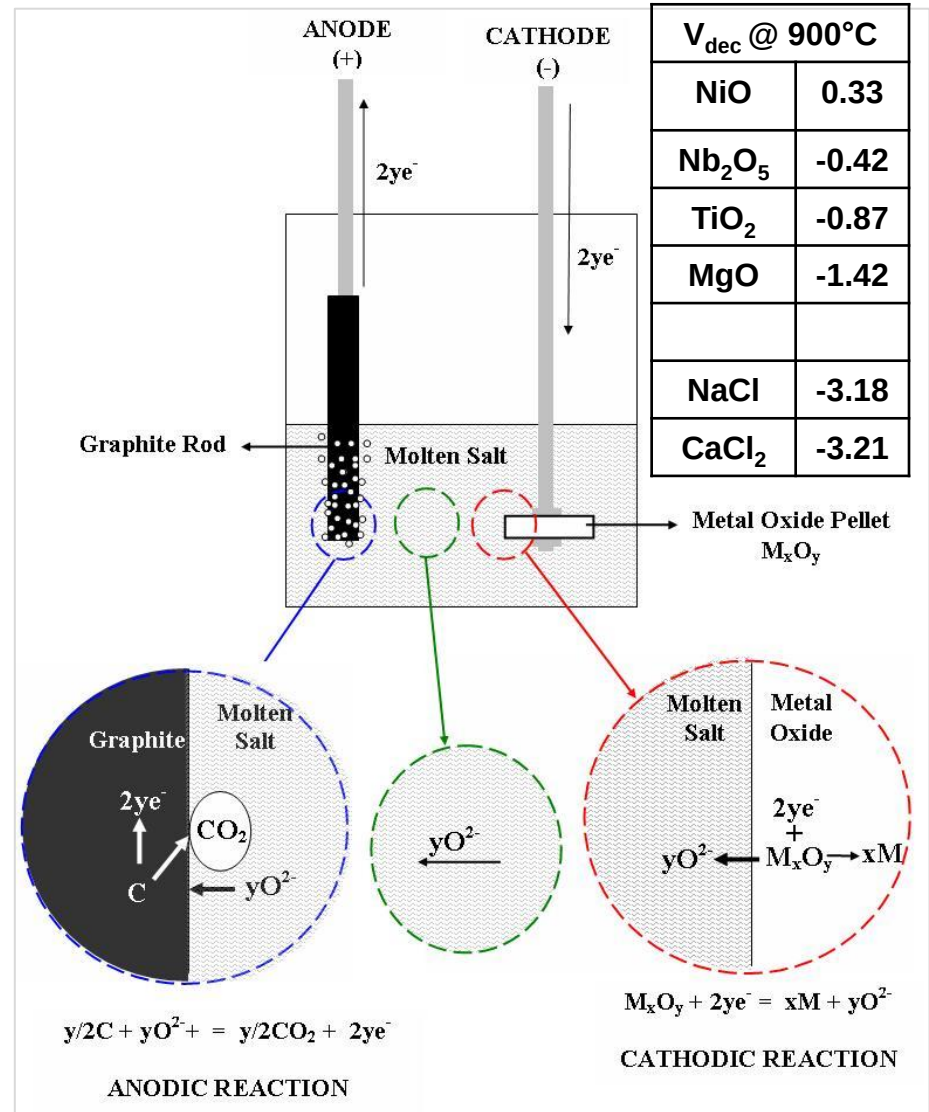
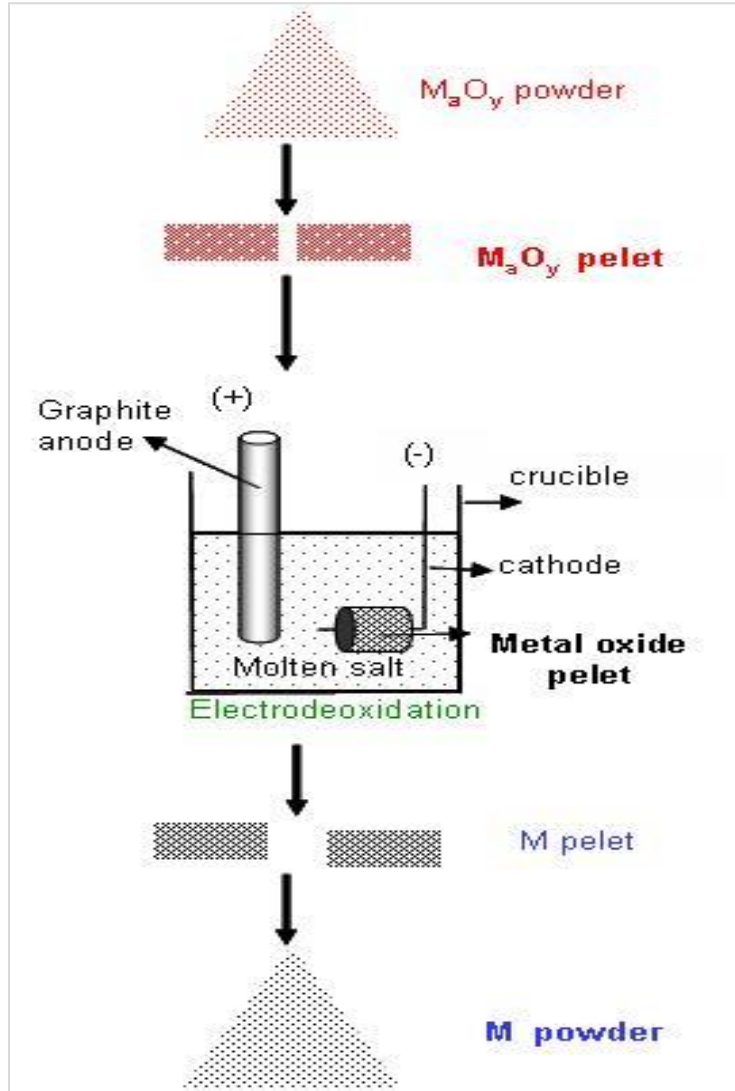


Flowchart for synthesis of hydrogen storage compounds
(Nanostructured HSA via Gas-Phase Synthesis)

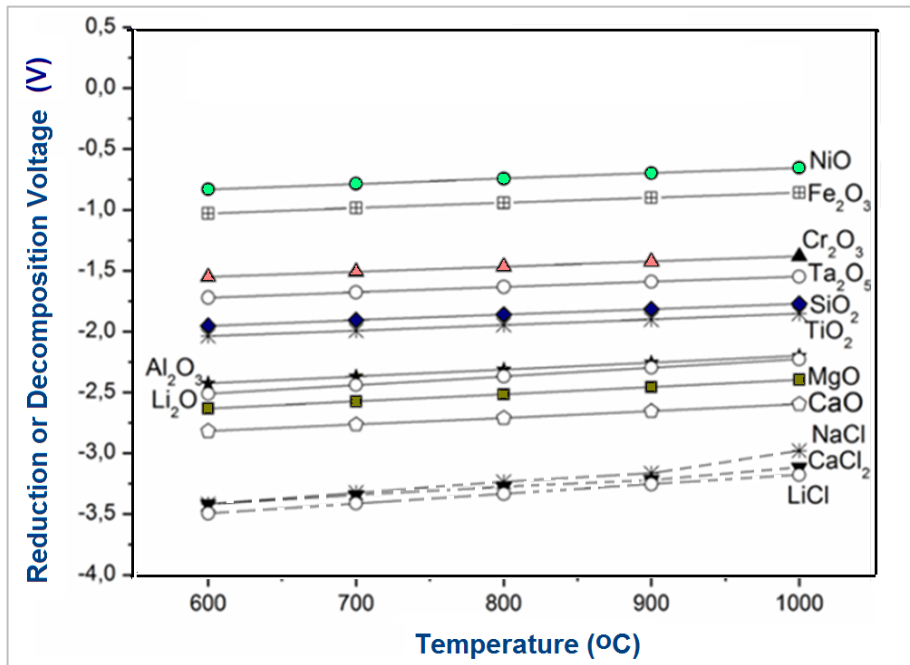


Flowchart for synthesis of hydrogen storage compounds
(Electro-reduction of Oxides to HSA)

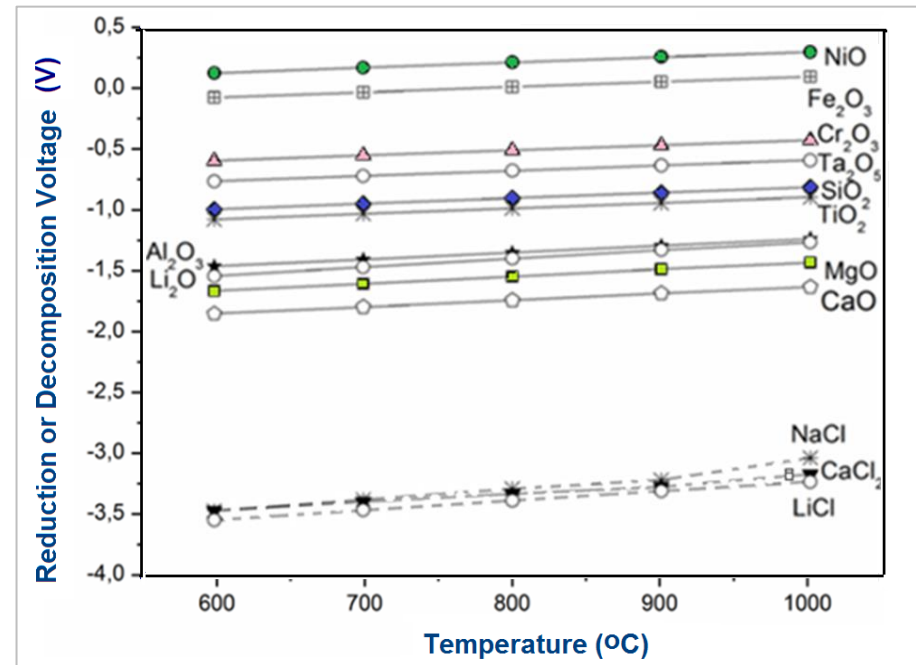
SOLID STATE ELECTROREDUCTION



Chen et al. Nature 407 (2000) 361–364.



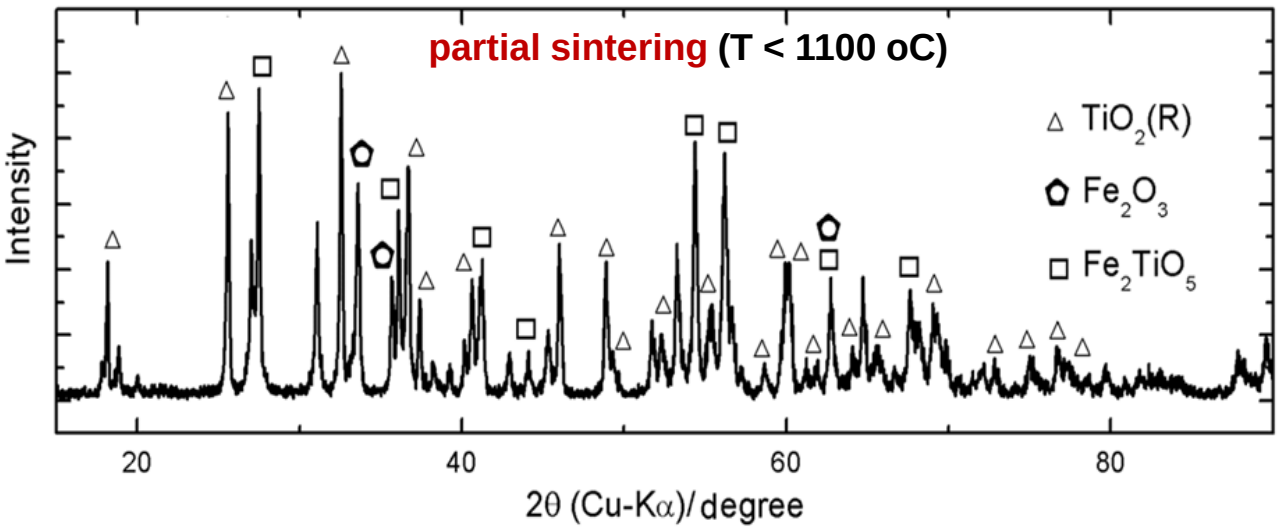
Reduction via O₂ evolution



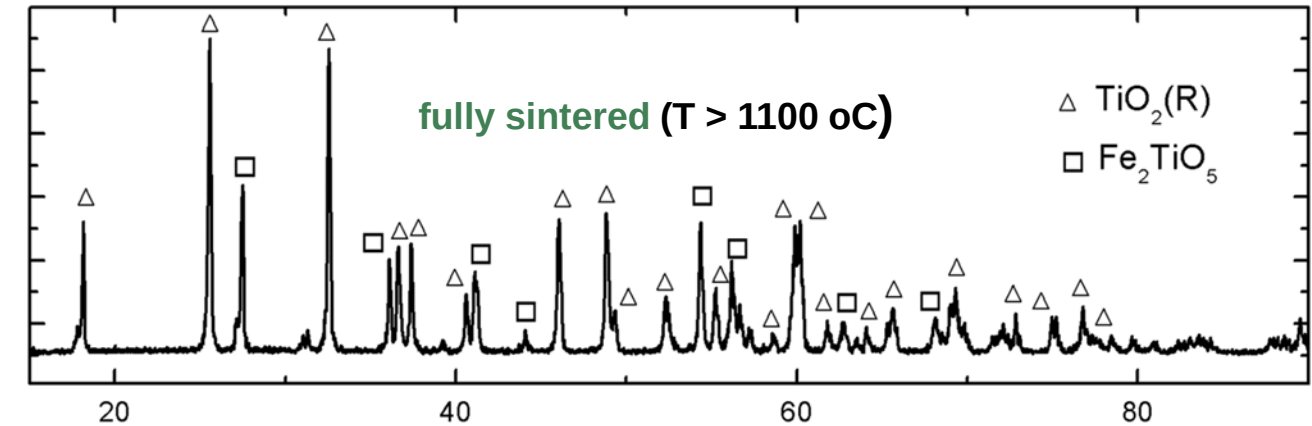
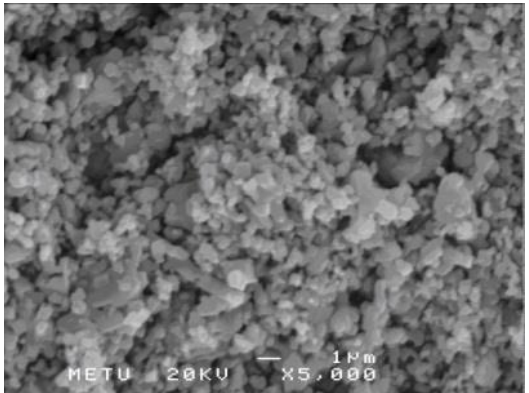
Reduction via CO₂ evolution

Standard reduction voltage (ΔEo) of selected oxides

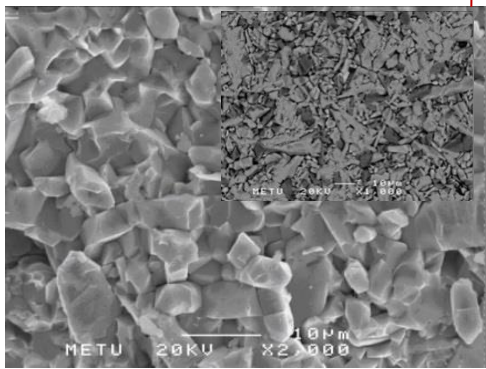
Synthesis of FeTi : Sintered Pellets



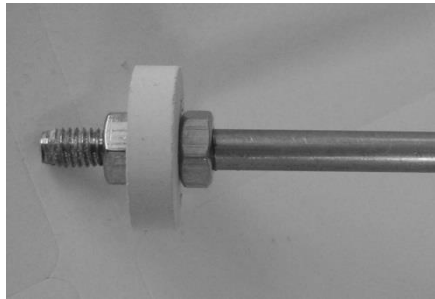
900 °C, 2h



1300 °C, 2h.



Synthesis of FeTi : Electroreduction of pellets



Before reduction



after reduction

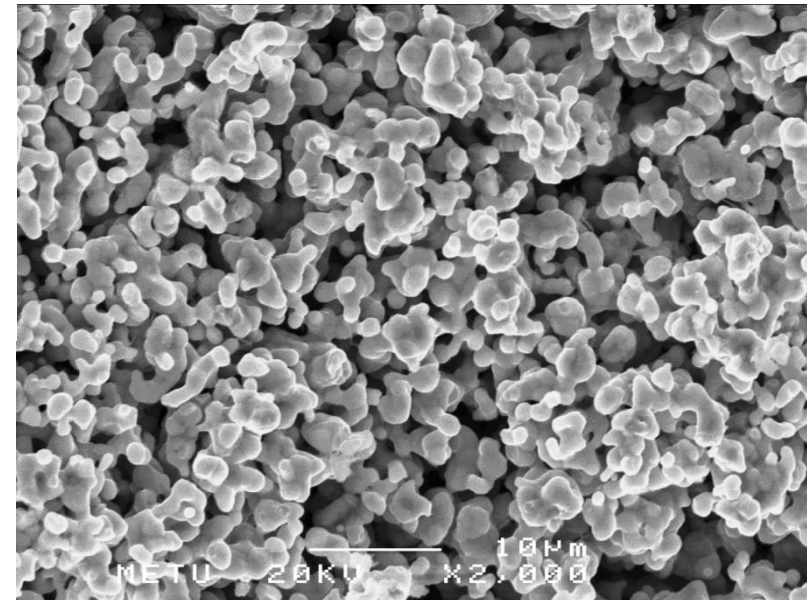
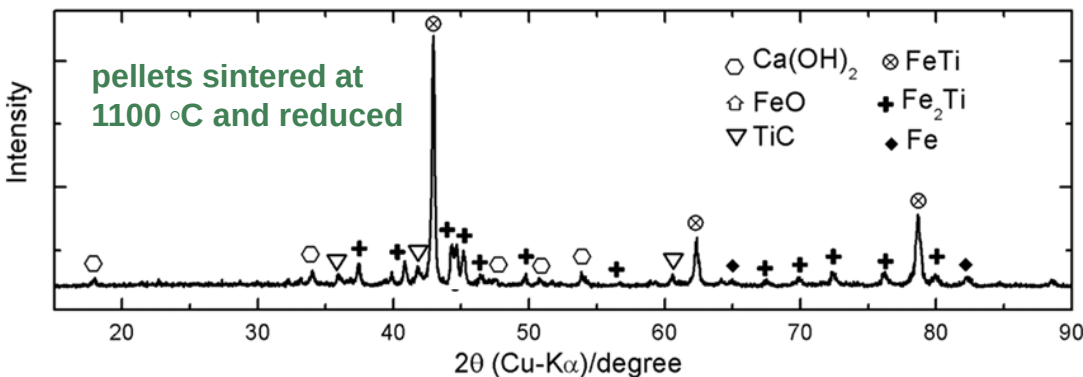
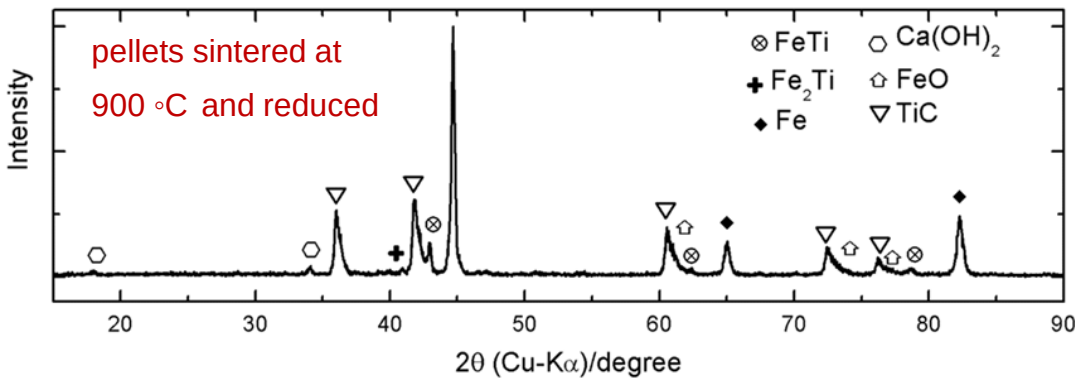
Pre-electrolysis

Stainless wire -graphite
3.0 V ; 6 hrs 900 oC)
Ar flow = 100-150 ml / min

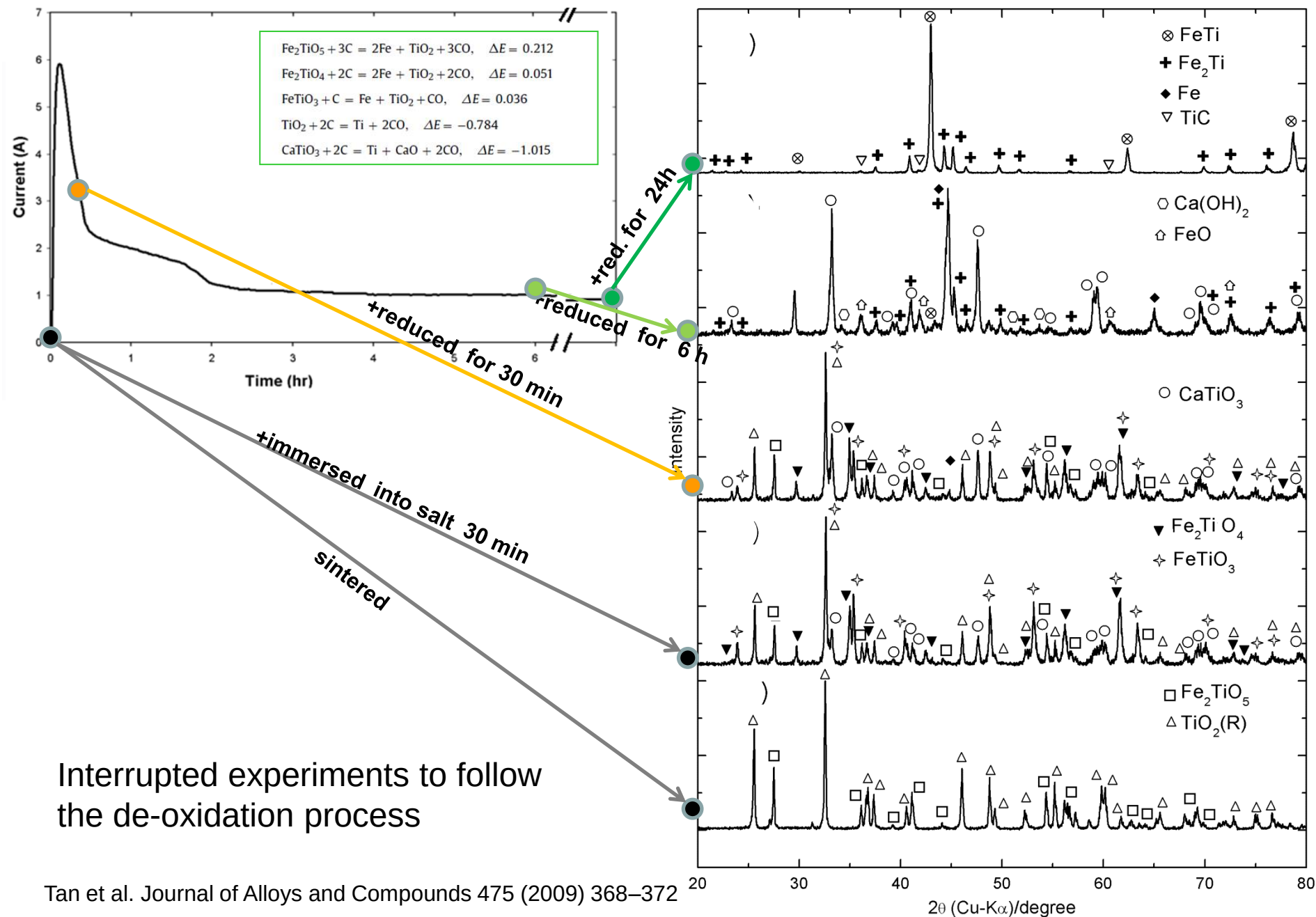


Electrodeoxidation

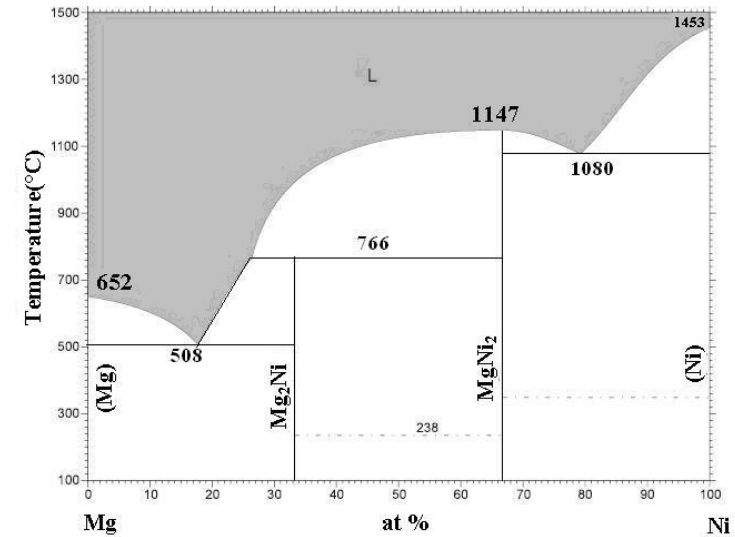
Sample -Graphite
(3.2 V ; 24 hrs 900)
Ar flow = 100-150 ml/min



FeTi, obtained from deoxidation of mixed oxide pellet sintered at 1100°C.



Mg-Ni system

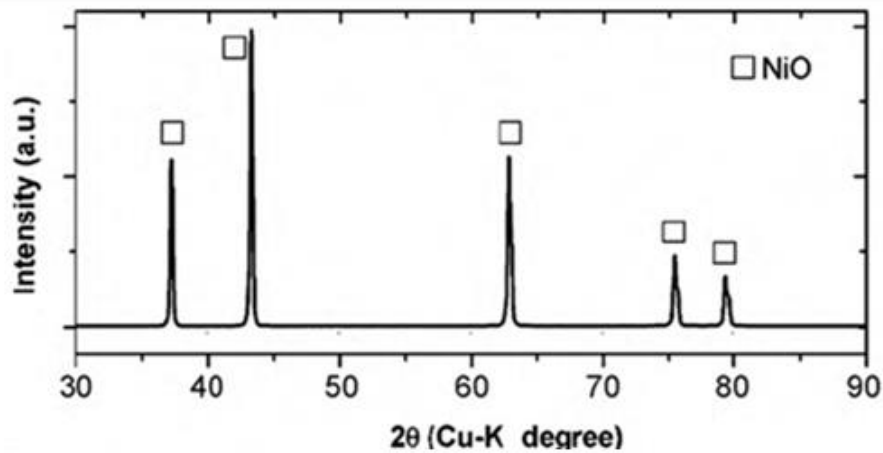


Mg-Ni Phase diagram

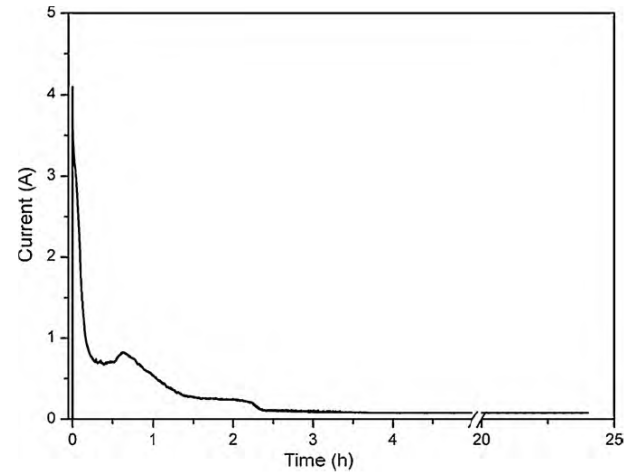
Conditions of electro-reduction

System	Target Phase	Electrolyte	V applied	T (oC)	t hr	Porosity (%)
NiO	Ni	CaCl ₂ +NaCl	3.2	900 725 600	24	57
MgO:NiO=1:2	MgNi ₂	CaCl ₂ +NaCl	3.2	900 600	24	42
MgO:NiO= 2:1	Mg ₂ Ni	CaCl ₂ +NaCl	3.2 5.0	600	24	35
MgO	Mg	CaCl ₂ +NaCl	3.2 5.0	600	24	35

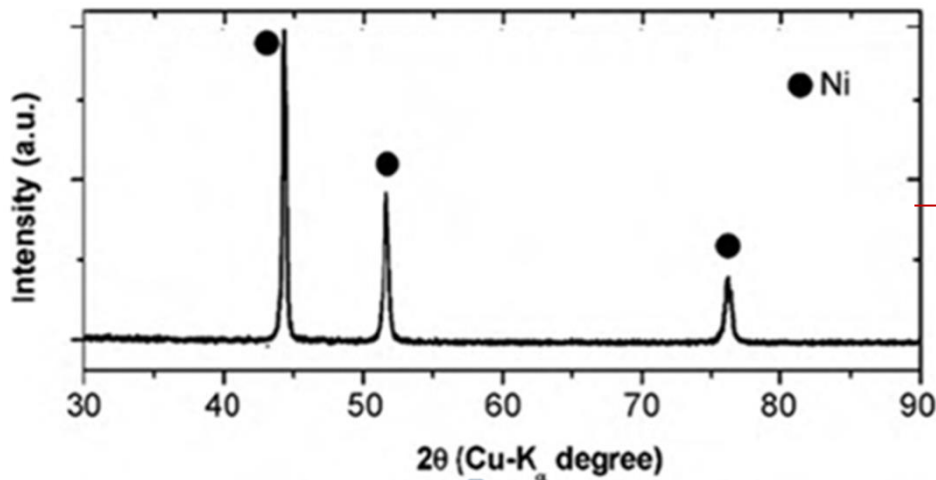
Synthesis Mg-Ni system : Ni synthesis



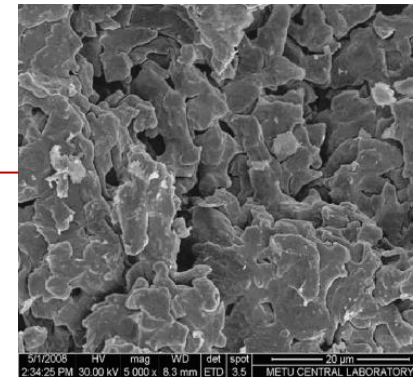
NiO pellet sintered at 1200 °C for 6 h



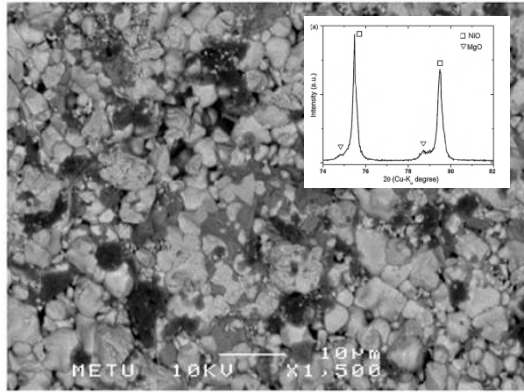
Current-time data collected during electroreduction



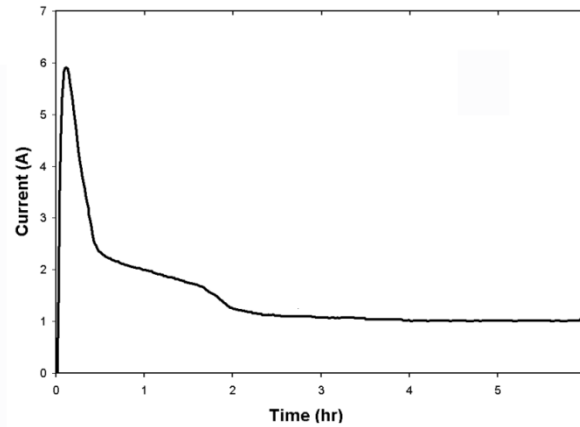
Electroreduced at 900 °C for 24 h (3.2 V).



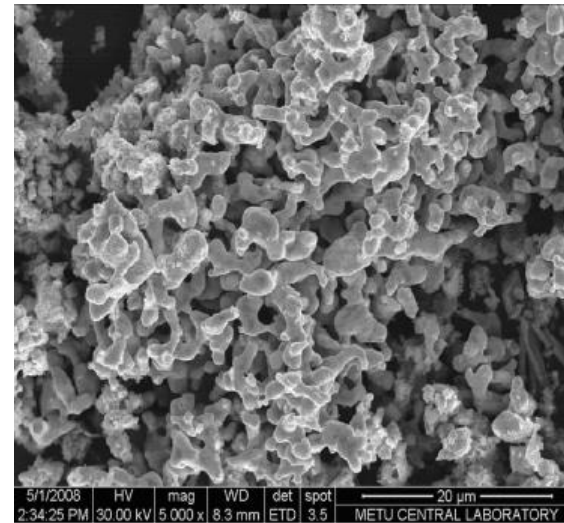
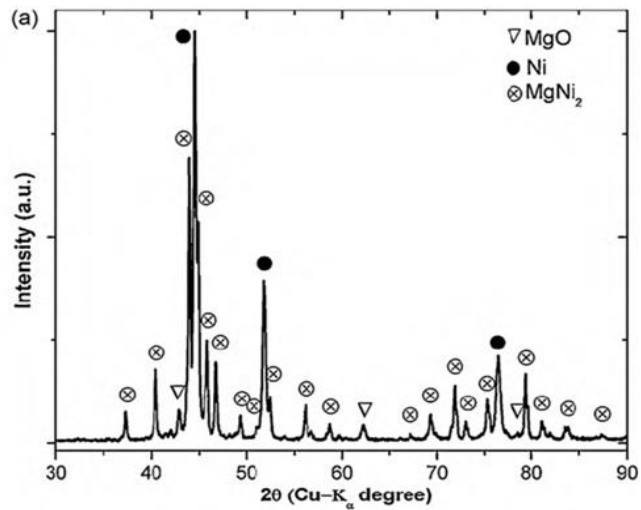
Synthesis in Mg-Ni system : MgNi_2



MgO:NiO = 1:2 pellet sintered at 1200°C for 6 h

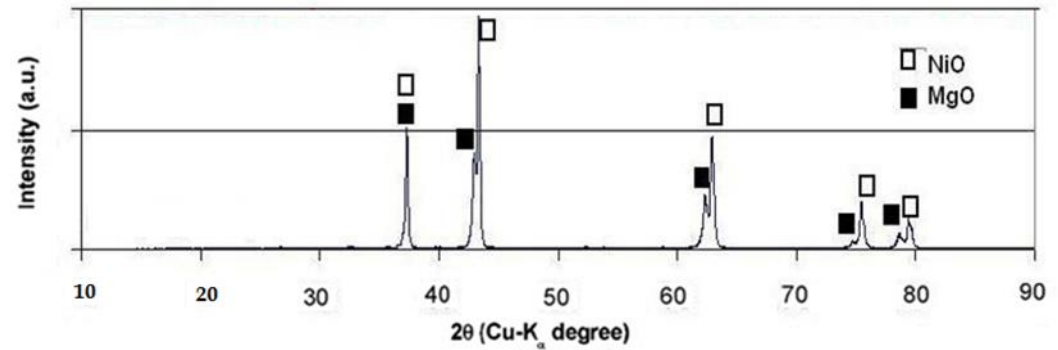
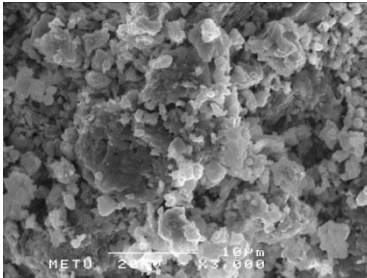


Current-time data collected during electroreduction at 900 °C 3.2 V

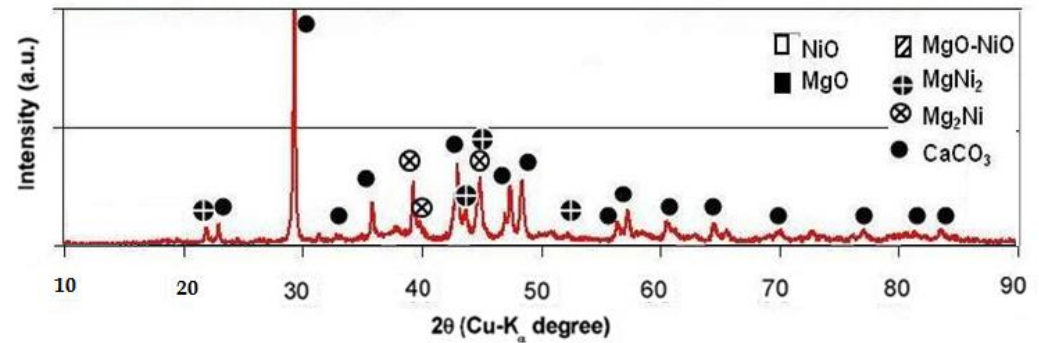
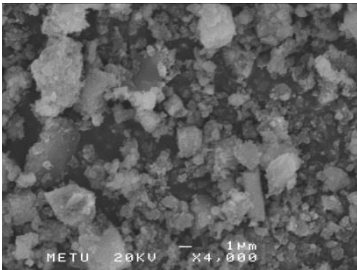


Electroreduced at 900 °C for 24 h (3.2 V).

Synthesis in Mg-Ni system : Mg₂Ni



MgO:NiO = 2:1 pellet sintered at 1200°C for 6 h

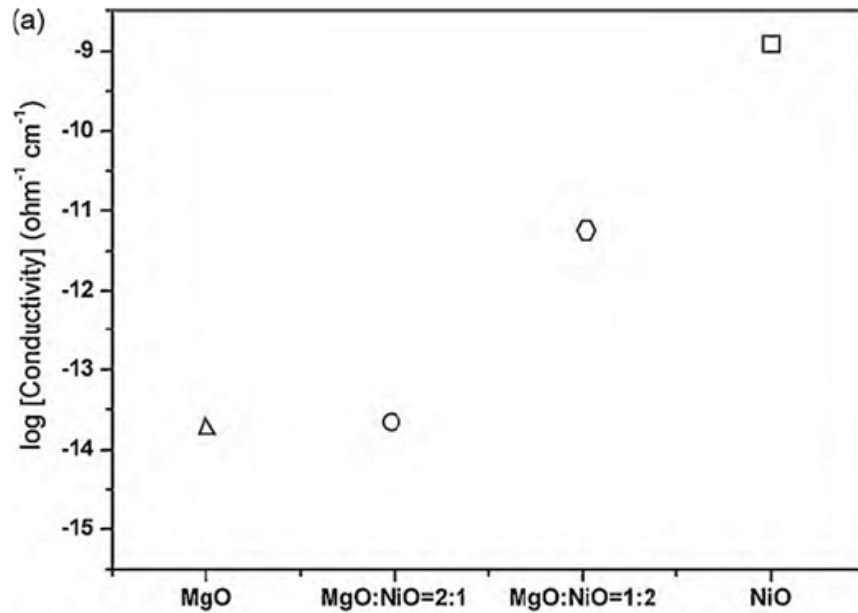


Electroreduced at 600 °C for 24 h (5 V).

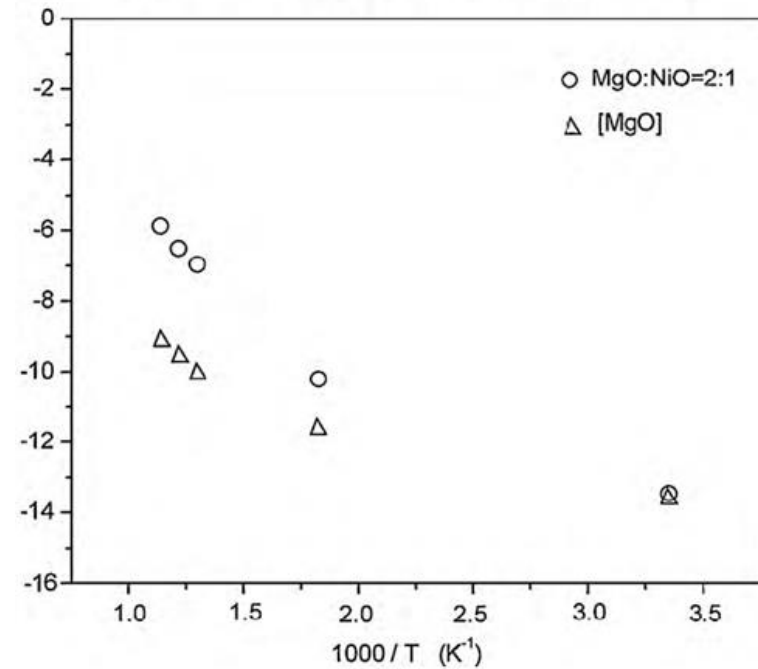
Tan et al Journal of Alloys and Compounds 504 (2010) 134–140

Standard reduction potentials at 900°C, 725°C and 600°C.

Reaction	E° (V)		
	600 °C	725 °C	900 °C
$NiO_{(s)} + 0.5C_{(s)} = Ni_{(s)} + 0.5CO_{2(g)}$	+0.20	+0.25	+0.33
$MgO_{(s)} + 2NiO_{(s)} + 3C_{(s)} = MgNi_{2(s)} + 3CO_{(g)}$	-0.36	-0.24	-0.09
$2MgO_{(s)} + 4NiO_{(s)} + 3C_{(s)} = 2MgNi_{2(s)} + 3CO_{2(g)}$	-0.32	-0.26	-0.18
$2MgO_{(s)} + NiO_{(s)} + 3C_{(s)} = Mg_2Ni_{(s,l)} + 3CO_{(g)}$	-0.98	-0.86	(-0.70)
$4MgO_{(s)} + 2NiO_{(s)} + 3C_{(s)} = 2Mg_2Ni_{(s,l)} + 3CO_{2(g)}$	-0.94	-0.87	(-0.78)
$MgO_{(s)} + C_{(s)} = Mg_{(s,l)} + CO_{(g)}$	-1.64	(-1.52)	(-1.34)
$MgO_{(s)} + 0.5C_{(s)} = Mg_{(s,l)} + 0.5CO_{2(g)}$	-1.60	(-1.53)	(-1.43)
$CaO_{(s)} + C_{(s)} = Ca_{(s,l)} + CO_{(g)}$	-1.83	-1.71	(-1.52)
$CaO_{(s)} + 0.5C_{(s)} = Ca_{(s,l)} + 0.5CO_{2(g)}$	-1.79	-1.72	(-1.63)
$NaCl_{(l)} = Na_{(l)} + 0.5Cl_{2(g)}$	(-3.37)	(-3.28)	(-3.16)
$CaCl_{2(l)} = Ca_{(s,l)} + Cl_{2(g)}$	-3.40	-3.32	(-3.21)



Electrical conductivities of MgO,
MgO:NiO = 2:1, MgO:NiO = 1:2 and
NiO at room temperature



Conductivity values as a function of
temperature up to 600 °C for MgO and
MgO:NiO = 2:1.

Volume change upon reduction

Material	Molecular weight g/mol)	Density g/cm ³	Normalized Molar volume cm ³ /mol	Change in volume %
MgO	40.30	3.58	11.26	+34.9
Mg	24.31	1.60	15.19	-
NiO	74.69	6.67	11.20	-36.33
Ni	58.69	8.23	7.13	
Fe ₂ O ₃	159.69	5.24	15.24	-50.18
Fe	55.85	7.36	7.59	-
TiO ₂	79.87	4.25	18.79	-40.72
Ti	47.87	4.29	11.14	-

Conclusions

Conventional processing of hydrogen storage alloys involve an excessive number of steps which contributes to their high cost;

- Electroreduction is one–step process that converts the oxide(s) directly into the product in particulate form (1-3 μm).
- For successful conversion, It is necessary to sinter the oxides fully while keeping an acceptable level of porosity.
- FeTi was successfully synthesized when sintered at $>1100^\circ\text{C}$ and reduced in CaCl_2 at 3.2 V at 900°C .
- The method could be used for many compounds. This is except for Mg and Mg rich compounds which have a peculiar property of volume expansion upon reduction.
- Electroreduction would give only equilibrium structures due to the high temperature involved. For non-equilibrium structure post-processing might be necessary via mechanical milling or gas-phase synthesis,

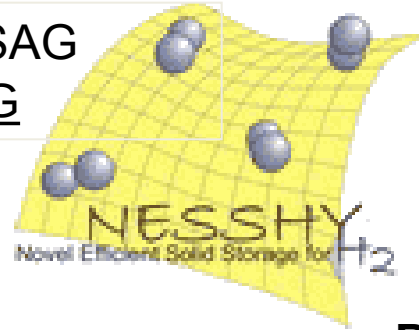
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- Burak Aktekin
- Ezgi Onur
- Seda Oturak
- Cavit Eyövge

Support

DPT
DPT-ÖYP
DPT-YUUP: "Hidrojen Gazı
Üretimi, Depolanması ve Yakıt
Hücrelerinde Elektrik Elde
Edilmesi"

TÜBİTAK- MİSAG
TÜBİTAK-MAG



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- Taylan Örs
- Çağla Özgüt
- Rabia Ölmez



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